


```
# Remove outliers
df = df[df['Magnitude'] > 0]
df = df[df['Depth'] > 0]

print("Clean data shape:", df.shape)
print("\nMagnitude range:", df['Magnitude'].min(), "to", df['Magnitude'].max())
print("Depth range:", df['Depth'].min(), "to", df['Depth'].max())
df.describe()
```

Clean data shape: (23239, 4)

Magnitude range: 5.5 to 9.1

Depth range: 0.02 to 700.0

	Latitude	Longitude	Depth	Magnitude	
count	23239.000000	23239.000000	23239.000000	23239.000000	
mean	1.392937	39.702321	71.294789	5.882613	
std	29.930594	125.766185	122.954923	0.423887	
min	-77.080000	-179.997000	0.020000	5.500000	
25%	-18.718500	-76.506600	15.000000	5.600000	
50%	-3.672000	106.075000	33.000000	5.700000	
75%	25.002850	145.285500	54.750000	6.000000	
max	86.005000	179.998000	700.000000	9.100000	

```
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
```

```
# Magnitude distribution
```

```
axes[0].hist(df['Magnitude'], bins=30, color='#FF6B35', edgecolor='black')
```

```
axes[0].set_title('Earthquake Magnitude Distribution')
```

```
axes[0].set_xlabel('Magnitude')
```

```
axes[0].set_ylabel('Count')
```

```
# Depth vs Magnitude
```

```
axes[1].scatter(df['Depth'], df['Magnitude'],
                alpha=0.3, color='#2196F3', s=1)
```

```
axes[1].set_title('Depth vs Magnitude')
```

```
axes[1].set_xlabel('Depth (km)')
```

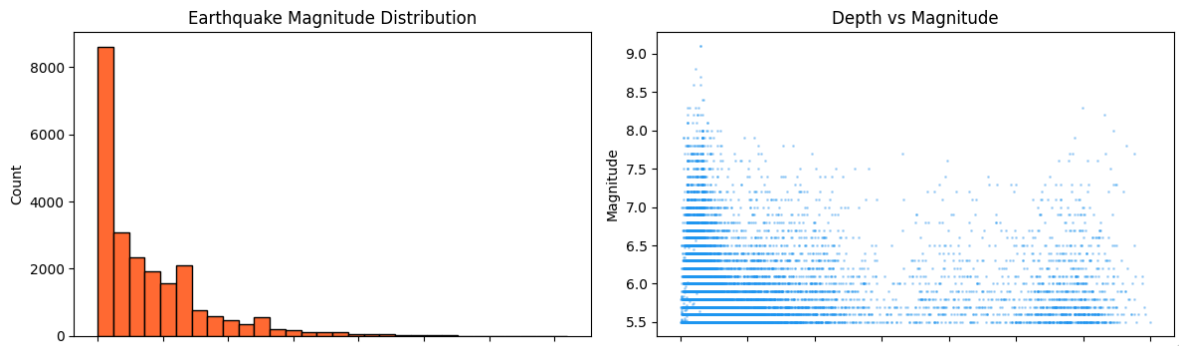
```
axes[1].set_ylabel('Magnitude')
```

```
plt.tight_layout()
```

```
plt.savefig('earthquake_analysis.png')
```

```
plt.show()
```

```
print("Saved!")
```



```
# Features and target
X = df[['Latitude', 'Longitude', 'Depth']]
y = df['Magnitude']

# Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Scale
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Train
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train_scaled, y_train)

# Evaluate
y_pred = model.predict(X_test_scaled)
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print(f"Mean Absolute Error: {mae:.3f}")
print(f"R2 Score: {r2:.3f}")
print(f"Model predicts magnitude within ±{mae:.2f} on average")
print("Model trained successfully!")
```

```
Mean Absolute Error: 0.329
R2 Score: -0.096
Model predicts magnitude within ±0.33 on average
Model trained successfully!
```

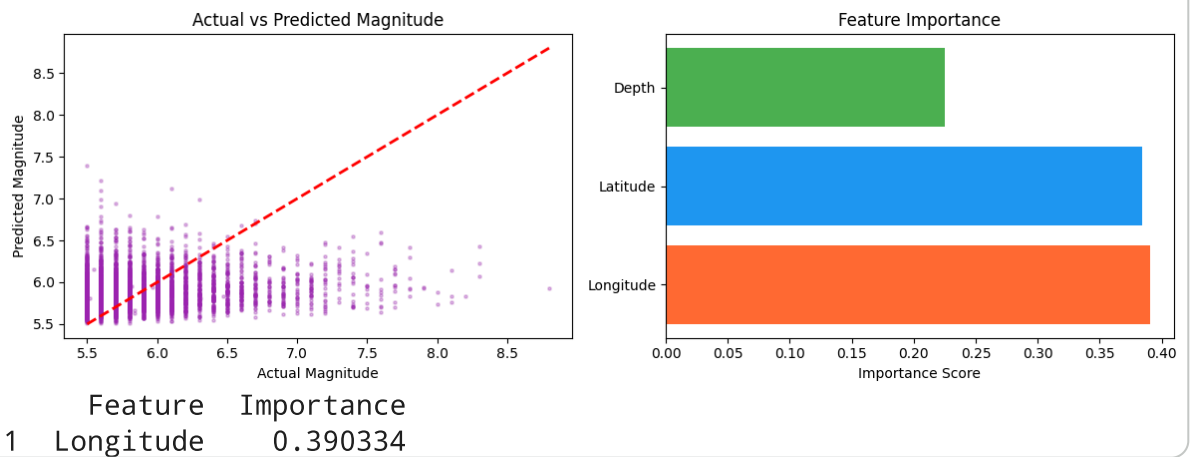
```
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

# Actual vs Predicted
axes[0].scatter(y_test, y_pred, alpha=0.3, color='#
axes[0].plot([y_test.min(), y_test.max()],
              [y_test.min(), y_test.max()],
              'r--', linewidth=2)
axes[0].set_title('Actual vs Predicted Magnitude')
axes[0].set_xlabel('Actual Magnitude')
axes[0].set_ylabel('Predicted Magnitude')
```

```
# Feature Importance
feat_imp = pd.DataFrame({
    'Feature': ['Latitude', 'Longitude', 'Depth'],
    'Importance': model.feature_importances_
}).sort_values('Importance', ascending=False)

axes[1].barh(feat_imp['Feature'], feat_imp['Importance'],
              color=['#FF6B35', '#2196F3', '#4CAF50'])
axes[1].set_title('Feature Importance')
axes[1].set_xlabel('Importance Score')

plt.tight_layout()
plt.savefig('earthquake_results.png')
plt.show()
print(feat_imp)
```



Start coding or [generate](#) with AI.

```
def predict_earthquake(latitude, longitude, depth):
    input_data = scaler.transform([[latitude, longitude, depth]])
    magnitude = model.predict(input_data)[0]

    if magnitude >= 7.0:
        level = "🔴 MAJOR – Serious damage expected"
    elif magnitude >= 5.0:
        level = "🟡 MODERATE – Some damage possible"
    else:
        level = "🟢 MINOR – Little to no damage"

    print(f"Predicted Magnitude: {magnitude:.2f}")
    print(f"Severity: {level}")

# Test: Japan 2011 Earthquake location
print("Testing Japan 2011 region:")
predict_earthquake(38.322, 142.369, 32)

print("\nTesting shallow depth earthquake:")
predict_earthquake(35.0, 25.0, 5)
```

Testing Japan 2011 region:
Predicted Magnitude: 6.74

Severity: MODERATE – Some damage possible

Testing shallow depth earthquake:

Predicted Magnitude: 5.78

Severity: 🟡 MODERATE – Some damage possible

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:273:
  warnings.warn(
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:273:
  warnings.warn(
```